

Economic Momentum and Currency Returns

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Abstract

Past trends in fundamentals linked to economic activity and inflation predict currency returns. We find that a trading strategy that goes long currencies with strong economic momentum and short currencies with weak economic momentum exhibits an annualized Sharpe ratio of 0.70 and yields a significant alpha when controlling for standard carry, momentum, and value strategies. The economic momentum strategy subsumes the alpha of carry trades, suggesting that differences in past economic trends capture cross-country differences in carry.

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1 Introduction

Present-value models suggest that the exchange rate can be written as a function of current and expected fundamentals. However, ever since the seminal work of [Meese and Rogoff \(1983\)](#), researchers have struggled to find support for these models, as macro fundamentals have problems predicting exchange rates out of sample. Rather than using macro fundamentals, researchers have shown cross-sectional and time-series predictability of currency returns in the form of carry, momentum, and value strategies. However, it is still widely debated what economic risks, if any, these strategies represent.¹

We find that macro fundamentals do predict currency returns. Past trends, measured over 1–60 months, in fundamentals linked to economic activity and inflation predict currency returns. A strategy that goes long currencies in countries with relatively strong economic momentum and short currencies in countries with relatively weak economic momentum exhibits an annualized Sharpe ratio of 0.70 over the 1976–2017 period. That is, for countries that have experienced relatively strong (weak) economic trends in the past, their currencies are associated with high (low) expected returns. Regressing the strategy returns on the returns of standard carry, momentum, and value strategies yields a statistically significant alpha, representing 58% of the average return of the economic momentum strategy. Hence, less than half of the average return is explained by common currency strategies. Introducing the economic momentum strategy into an ex post mean-variance efficient portfolio of carry, momentum, and value considerably reduces the portfolio weights of existing strategies while

¹The literature on carry, momentum, and value strategies in currency markets is vast. For example, carry trade returns have been linked to aggregate consumption and macroeconomic risks ([Colacito, Croce, Gavazzoni, and Ready, 2018](#); [Hoffmann and Studer-Suter, 2017](#); [Lustig and Verdelhan, 2007](#); [Zviadadze, 2017](#)), crash and downside risks ([Brunnermeier, Nagel, and Pedersen, 2009](#); [Chernov, Graveline, and Zviadadze, 2018](#); [Daniel, Hodrick, and Lu, 2017](#); [Farhi, Fraiberger, Gabaix, Ranciere, and Verdelhan, 2015](#); [Jurek, 2014](#); [Lettau, Maggiori, and Weber, 2014](#)), peso problems ([Burnside, Eichenbaum, Kleshchelski, and Rebelo, 2011](#)), and volatility risk ([Menkhoff, Sarno, Schmeling, and Schrimpf, 2012a](#)), while [Lustig, Rousanov, and Verdelhan \(2011\)](#) construct a carry-trade risk factor. Studies of momentum and value strategies include [Menkhoff, Sarno, Schmeling, and Schrimpf \(2012b, 2017\)](#), [Asness, Moskowitz, and Pedersen \(2013\)](#), and [Barroso and Santa-Clara \(2015a\)](#).

allocating a weight of 45% to economic momentum, suggesting the strategy potentially offers a different source of returns for a currency investor. Moreover, we find that returns based on economic momentum subsume the alpha of carry-trade strategies. This suggests that cross-country differences in interest rate differentials are captured by differences in past economic trends, where countries that rank high (low) in terms of carry are countries that have experienced high (low) economic activity and inflation in the past.

We use simplicity and robustness as a guide throughout our analysis. First, we construct country-level indices of economic activity and inflation through an equal-weighted average of growth rates in the underlying fundamental variables. The economic-activity indices are comprised of growth rates in industrial production, retail sales, and the inverse of unemployment while the inflation indices build on growth rates in consumer and producer price indices. Increases in these indices are associated with positive economic momentum. Second, we measure economic trends in the form of log changes in each index for lookback periods of 1–60 months. Third, we form dollar-neutral trading strategies for each index and lookback period, and for a combination of trends. We find that the cross-section of past economic trends significantly predicts excess returns up to a horizon of 12 months. While past trends in fundamentals all contain complementary information about expected returns, the strongest contribution to performance comes from long-term trends over the past three to five years. Long-term trends in fundamentals also most strongly capture the carry-trade alpha.

While the economic momentum strategy subsumes the alpha of carry trades, carry trades cannot fully explain the returns of the economic momentum strategy. The subsumption of carry is also evident when introducing the economic momentum strategy in an ex post mean-variance efficient portfolio of carry, value, and momentum as it renders the portfolio weight of carry statistically insignificant. Moreover, we consider the portfolio weights of currencies for the economic momentum strategy and the carry trade and find that they display different dynamics. For example, the portfolio weights for the economic momentum strategy and the

carry trade have different signs 38% of the time, measured across currencies. Compared to the portfolio weights of the momentum and value strategies, the portfolio weights of the economic momentum strategy have different signs approximately 50% of the time. Hence, investing according to past trends in fundamentals seems to represent an additional source of returns.

Interestingly, traditional momentum strategies (based on the exchange rate, that is, the price of one currency in terms of another currency) cannot explain the returns of the economic momentum strategy, suggesting that price and economic momentum strategies are distinct. In addition, results from predictive regressions suggest that economic momentum predicts currency returns up to 12 months ahead while the predictive power of price momentum diminishes beyond one month. These results relate to [Novy-Marx \(2015\)](#) and [Huang, Zhang, and Zhou \(2018\)](#), who find that in the stock market the returns based on momentum in firm fundamentals cannot be explained by the returns based on momentum in prices. Moreover, [Huang, Zhang, and Zhou \(2018\)](#) find that returns from fundamental momentum are more persistent over time.

We consider a rich set of robustness checks including different subsets of countries, time periods, and methods of portfolio construction. In particular, we construct time-series strategies based on economic momentum in which a long (short) position is taken in a foreign currency if the economic trend in that particular country is stronger (weaker) than that of the home country, the United States. The time-series strategies are not dollar neutral and can be either long or short the US dollar (USD). We find that a time-series strategy based on economic momentum generates a Sharpe ratio of 0.59 while yielding a statistically significant alpha when controlling for time-series benchmark strategies. The alpha accounts for 72% of the average return of the time-series combo. Hence, the profitability of the economic momentum strategy seems to be both a cross-sectional and a time-series phenomenon.

Our findings imply that fundamentals do matter for exchange rates but raise the question

of why economic momentum predicts currency returns. Several possible explanations could be considered, but one particular explanation that seems applicable is that of monetary policy and the role of [Taylor \(1993\)](#) rules, relating short-term interest rates and monetary policy to economic activity and inflation. For example, [Clarida and Waldman \(2008\)](#) and [Molodtsova and Papell \(2009\)](#) find Taylor-rule models to have significant predictive power for exchange rates, even though [Engel, Lee, Liu, Liu, and Wu \(2018\)](#) question the ability of a Taylor-rule model to produce superior out-of-sample forecasts. In addition, provided that the monetary authorities of a country adhere to a Taylor-type rule, then strong economic conditions and inflation in the past likely induced a tightening of monetary policy. In this case, countries with currently high (low) carry likely experienced positive (negative) past trends in economic fundamentals. Cross-sectional carry can therefore be captured by relative trends in fundamentals, which explains why the alpha from carry trades is subsumed by the economic momentum strategy.

Moreover, our finding that economic momentum predicts currency excess returns up to 12 months ahead suggests that models of delayed overshooting could shed light on our findings. [Eichenbaum and Evans \(1995\)](#) find that contractionary shocks to US monetary policy lead to an appreciation of the USD over an extended period of time. Extending the same reasoning across currencies, it seems likely that countries with strong economic momentum would be subject to stronger contractionary monetary policy shocks than countries with weak economic momentum. A model of delayed overshooting would then predict a prolonged period of high currency returns in countries with strong economic momentum compared to countries with weak economic momentum, which would be subject to a less contractionary or even expansionary monetary policy. Moreover, the combination of strong economic momentum, higher interest rates (e.g., via a Taylor-type rule), and an increase in demand for a particular currency from investors with delayed portfolio responses could generate a delayed overshooting effect along the lines of [Bacchetta and van Wincoop \(2010\)](#). However, we cannot rule

out alternative explanations for our findings.

Our findings are also related to a recent literature on the relationship between currency returns and fundamentals. For example, [Engel and West \(2005\)](#), [Engel, Mark, and West \(2007\)](#), and [Sarno and Schmeling \(2014\)](#) all find that fundamentals and expectations of fundamentals matter for exchange rates. Moreover, currency returns have been linked to countries' external imbalances ([Gourinchas and Rey, 2007](#); [Della Corte, Sarno, and Sestieri, 2012](#); [Della Corte, Riddiough, and Sarno, 2016](#)), macroeconomic uncertainty ([Berg and Mark, 2018](#); [Della Corte and Krecetovs, 2017](#)), sovereign risk ([Della Corte, Sarno, Schmeling, and Wagner, 2018](#)), factor model forecasts ([Engel, Mark, and West, 2015](#)), unemployment ([Nucera, 2017](#)), and the output gap ([Colacito, Riddiough, and Sarno, 2018](#)). However, these studies do not directly consider trading strategies based on short- and long-term trends in a broader set of fundamentals as we do for our economic momentum strategy.

Section [2](#) describes the data while Section [3](#) describes how we measure economic momentum from trends in fundamental variables and how we form portfolios. Section [4](#) presents and discusses the performance of the economic momentum strategy. Section [5](#) concludes.

2 Data

2.1 Exchange rates

We retrieve daily data on spot and one-month forward exchange rates from *Datastream* for the period from January 1976 to March 2017. The main results refer to developed markets, comprised of the following twenty-one countries: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, and the United States, as well as the Eurozone. We also consider a set of currencies comprising the G10 as well as a set of emerging market countries. The Online Appendix lists all countries, to which set of

countries they belong, and over which period data are available. All currencies are expressed in USD per unit of foreign currency.² An increase in the exchange rate of currency c at time t , $S_{c,t}$, implies appreciation of the foreign currency and depreciation of the USD. We assume the USD to be the home currency, the excess return on investing in foreign currency c via a forward contract, $F_{c,t}$, being denoted $R_{c,t+1} = (S_{c,t+1} - F_{c,t})/F_{c,t}$. Currencies now in the Eurozone are used only up to the date of entering the Eurozone or up to the date the currency was fixed against the Euro, in most cases December 31, 1998. The Online Appendix contains a more detailed description of data (including *Datastream* mnemonics for the spot and forward exchange rates series) and data cleaning.

2.2 Fundamental variables

We collect data on five fundamental variables: Industrial production, retail sales, unemployment, consumer prices, and producer prices. All fundamental data are collected from the statistical database of the OECD. Using the five fundamental variables, we construct country-level indices that measure economic activity and inflation.³ We use the simplest possible construction in the form of an equal-weighted average of growth rates in the underlying variables as to make our findings as intuitive and transparent as possible. More specifically, the economic-activity index is constructed through an equal-weighted average of growth rates in industrial production, retail sales, and the inverse of unemployment. Likewise, the inflation index is constructed as an equal-weighted average of growth rates in consumer and

²We use forward contracts against the British pound up to February 1, 2007, as they provide the longest available history, after which we use forward contracts against the USD.

³It is common in the asset-pricing literature to separate the effects of economic activity and inflation on asset prices; for example [Ang and Piazzesi \(2003\)](#) introduce economic growth and inflation in a no-arbitrage term structure model, [Piazzesi and Schneider \(2006\)](#) model consumption growth and inflation to price nominal bonds, [Andersen, Bollerslev, Diebold, and Vega \(2003, 2007\)](#) study the separate effects of news related to economic activity and inflation for the process of price discovery, [Beber, Brandt, and Luisi \(2015\)](#) construct timely measures of economic activity and inflation, and [Zviadadze \(2017\)](#) studies the sensitivities of different currency baskets to shocks to consumption growth, inflation, and interest rates.

producer price indices.⁴ Increases in the indices are interpreted as improved economic conditions. We verify that all variables underlying the indices indeed are positively related to the growth in industrial production. Results of these panel regressions are reported in the Online Appendix. We use monthly data over the period from January 1976 to March 2017, but for a few countries and fundamental variables we use quarterly data (with repeated values) as monthly data are not available. The availability of fundamental variables varies across time and countries. Consequently, we use all available fundamental variables at a particular point in time when constructing the indices.

3 Economic momentum and portfolio construction

The main results are based on measuring trends using log growth rates in the fundamental indices. The construction of the trend measure is kept simple to minimize any data-mining concerns. For robustness, we have also considered trend measures based on linear time-trend regressions, quadratic time-trend regressions, and moving averages with very similar results.⁵ The objective is to measure various trend frequencies and avoid an arbitrary lookback period wherefore we consider lookback periods of 1–60 months. Restricting the lookback periods to multiples of 12 months (i.e., lookback periods of 12, 24, 36, 48, and 60 months) makes little difference to the main results. All trend measures are computed at the end of each month when portfolio rebalancing occurs.⁶

⁴The two inflation measures provide complementary information as they reflect price pressures in different stages of the supply chain. For example, so called cost-push inflation is likely to first be reflected in producer prices. Indeed, we find the average correlation between consumer and producer-price inflation to be low, 0.29, measured across countries.

⁵The literature on decomposing a time series into a trend and a cyclical component is vast and stretches from classical contributions such as [Beveridge and Nelson \(1981\)](#) to recent ones by, for example, [Hamilton \(2018\)](#). We have considered the simplest possible trend measures as to keep our procedure as practically viable and transparent as possible.

⁶The results are based on using three-month-lagged information when forming portfolios to guarantee that investors had access to relevant macro information when rebalancing and to address the fact that macro data are often published with a lag. However, such information lagging has a small impact on the final results. A three-month lag together with a minimum lookback period of one month means that the first

3.1 Momentum in economic variables

The trend measure z is based on log growth rates in the fundamental indices, defined as $z_{c,i,l,t} = \ln(X_{c,i,t}) - \ln(X_{c,i,t-l})$, where $X_{c,i,t-l}$ denotes for currency c the particular index i at time t for the lookback period l , which varies between one and 60 months. The investment signal is based on the magnitude of the growth rate, measuring the strength of the time trend, and standardizing the growth rate by past volatility makes little difference to the main results. An alternative to varying the lookback period between one and 60 months would be to use back-end and/or front-end smoothing (see, e.g., [Levine and Pedersen, 2016](#)).

3.2 Construction of economic momentum portfolios

Having described how we construct the trend measure, we now describe how portfolios are formed. Two dimensions must be considered when constructing portfolios, namely, the fundamental index and the lookback period. We construct a strategy for each index and for each lookback period. As there are two types of indices and 60 lookback periods, there are in total 120 different sub-strategies. We later aggregate these sub-strategies based on the fundamental index and the lookback period. We also consider a strategy that diversifies across all available sub-strategies.

At the end of each month, currencies are ranked according to the strength of the trend for a particular index and lookback period. We weight currencies according to their cross-sectionally ranked signal, similar to the method used by [Asness, Moskowitz, and Pedersen \(2013\)](#) and [Kojen, Moskowitz, Pedersen, and Vrugt \(2018\)](#). Every month, the rank-based weight for currency c using fundamental index i and for lookback period l at time t is defined

portfolio weights are formed in May 1976 (based on the growth rate in the fundamental indices between January and February 1976) and that the monthly returns for the trend strategies start in June 1976. We have also constructed the trend strategies using non-revised real-time macro data from OECD. The data are available from 1999 for our variables and the results are very similar to our main results.

as:

$$w_{c,i,l,t} = \kappa_t \left[\text{rank}(z_{c,i,l,t}) - \frac{1}{C_t} \sum_{c=1}^{C_t} \text{rank}(z_{c,i,l,t}) \right], \quad (1)$$

where $z_{c,i,l,t}$ denotes the investment signal for currency c using fundamental index i for lookback period l at time t , C_t denotes the number of available currencies at time t , and κ_t denotes a scaling factor ensuring that each sub-strategy invests one dollar on the long side and one dollar on the short side. Note that this ranked-based weighting makes the strategy dollar neutral. An advantage with the ranked-based weighting scheme is that it reduces the impact of measurement errors, revisions, or outliers in data, particularly in conjunction with the long lookback periods that we consider. For example, for an outlier to have a significant impact on the results, the outlier must alter long-term macro trends and the relative ranking among countries. In addition, the ranked-based approach potentially provides diversification as the investment then includes all assets in the universe compared to, for example, investing in the top and bottom quintiles of assets. The portfolio return at time $t + 1$ for fundamental index i and lookback period l is then:

$$R_{i,l,t+1} = \sum_{c=1}^{C_t} w_{c,i,l,t} R_{c,t+1}. \quad (2)$$

Using the sub-strategies, we form aggregate portfolios across indices and lookback periods. As the volatility of each sub-strategy is different, we weigh each sub-strategy by the inverse of its past volatility and scale the weights such that they sum to one. The past volatility is computed as the exponentially weighted moving average volatility of strategy returns based on daily spot returns, using a RiskMetrics lambda of 0.94.⁷ We later report results of individual portfolios for each index, aggregate portfolios using short-, medium-, or long-term trends, and finally a combined trend portfolio (“combo”) covering all sub-strategies.

⁷We use a “burn-in” period for the volatility estimation of three years. During that initial period, the sub-strategies are equally weighted.

While the portfolio weights in Equation (1) generates dollar-neutral portfolios, Section 4.8 considers portfolio weights that have a dollar exposure.

3.3 Construction of benchmark strategies

We construct a number of portfolios based on well-established currency strategies. We use these additional strategies as benchmarks when evaluating the performance of the trend strategies. Closely following the existing literature, we construct benchmark strategies in the forms of carry, momentum, and value strategies. All strategies are constructed using cross-sectional ranking, similar to Equation (1), and represent dollar-neutral portfolios as to match the construction of the trend strategies. All strategies use end-of-month rebalancing.

The carry portfolio is constructed by sorting countries according to their forward premia. The carry trade is often implemented by going long currencies that trade at a forward discount ($F_{c,t} < S_{c,t}$) and short currencies that trade at a forward premium ($F_{c,t} > S_{c,t}$).⁸ Such a portfolio will typically have exposure to the dollar. However, to be consistent with the dollar neutrality of the trend strategies, we instead rank currencies each month according to their forward premia. We build portfolios using the rank-based weighting:

$$w_{c,t} = \kappa_t \left[\text{rank}(S_{c,t} - F_{c,t}) - \frac{1}{C_t} \sum_{c=1}^{C_t} \text{rank}(S_{c,t} - F_{c,t}) \right], \quad (3)$$

where the long and short side of the strategy both have one dollar invested, implying a zero-cost and dollar-neutral portfolio.

Momentum and value portfolios are constructed similar to how [Asness, Moskowitz, and Pedersen \(2013\)](#) construct portfolios. The momentum portfolio ranks currencies according to their returns over the past 12 months. The strategy then goes long currencies with high

⁸Covered interest-rate parity implies that the forward premia can be expressed in terms of the difference between domestic (here US) and foreign one-month interest rates. The carry trade is then implemented by going long (short) currencies with interest rates higher (lower) than the domestic (here US) interest rate.

past returns and short currencies with low past returns. The value portfolio sorts currencies according to the negative of the log change in real exchange rates over the past 60 months. More specifically, the value signal for currency c in month t is defined as $(\pi_{t-60:t} - \pi_{c,t-60:t}) - \Delta s_{c,t-60:t}$, where $\pi_{t-60:t}$ denotes US log inflation over the last 60 months, $\pi_{c,t-60:t}$ denotes the corresponding foreign log inflation, and $\Delta s_{c,t-60:t}$ denotes the log depreciation of the USD over the last 60 months. The strategy then goes long high-value currencies and short low-value currencies.

The part of the value signal that measures foreign inflation, $(-\pi_{c,t-60:t})$, is in fact the negative of the trend signal that measures inflation over the past 60 months, with the exception that the trend strategy uses information from both consumer and producer prices while the value signal only uses consumer price inflation. We find that past inflation predicts currency returns positively which means that the inflation part of the value signal contributes negatively to the performance of the value strategy. In fact, when decomposing the value signal into an inflation part, $(\pi_{t-60:t} - \pi_{c,t-60:t})$, and a currency spot-return part, $-\Delta s_{c,t-60:t}$, we find that the profitability of the value signal comes entirely from the latter part.

We construct momentum and value portfolios using cross-sectional ranking as in Equation (3). [Asness, Moskowitz, and Pedersen \(2013\)](#) find that combinations of value and momentum strategies provide diversification benefits. Moreover, [Jordà and Taylor \(2012\)](#) find diversification benefits of combining carry trades with a value-based measure of exchange rates, and [Kroencke, Schindler, and Schrimpf \(2014\)](#) and [Barroso and Santa-Clara \(2015a\)](#) find large benefits of diversifying across carry, momentum, and value strategies. We will therefore also consider the economic momentum strategies from a portfolio investment perspective.

4 Performance

We now present and discuss the performance of the trend strategies. We report performance statistics for trend strategies based on each of the two fundamental indices, for strategies based on short-, medium-, and long-term trends, and for strategies diversified across indices and lookback periods. We then relate the trend strategies to the various benchmark strategies, run predictive panel regressions, and specifically relate the trend strategies to the carry strategy. We also consider diversification benefits for a currency investor. Finally, we discuss various robustness checks.

4.1 Performance of economic momentum

We construct portfolios for each of the two fundamental indices and for each lookback period. Figure 1 reports annualized Sharpe ratios for all sub-strategies. The figure shows that all sub-strategies have positive Sharpe ratios. The Sharpe ratios not only differ across indices but also across lookback periods for each index. The return correlations across indices and lookback periods are less than perfect. Consequently, an investor can achieve diversification benefits by forming an aggregate portfolio of sub-strategies. Combining investment signals in this manner is analogous to diversifying across predictive variables when making forecasts, which, for example, [Wright \(2008\)](#), [Della Corte, Sarno, and Tsiakas \(2009\)](#), and [Rapach and Zhou \(2013\)](#) have shown improves out-of-sample forecasting performance. See also [Timmermann \(2006\)](#) for a survey on forecast combinations.

We construct such aggregate portfolios for each of the two sub-combos and report performance statistics in Table 1. The combo based on economic activity has a Sharpe ratio of 0.39 with positive skewness while the inflation combo exhibits a higher Sharpe ratio of 0.66 coupled with negative skewness. The inflation combo displays higher kurtosis than the economic-activity combo. The cumulative returns of the two sub-combos are plotted in Fig-

ure 2. The correlation of returns between the two sub-combos is 0.20, suggesting potential diversification benefits of combining economic activity and inflation. Such benefits are exemplified by the financial crisis in which the average return of the economic-activity combo was positive over the period June 2007 to June 2009 whereas the inflation combo suffered a significant loss.

Building on the low correlation, we consequently construct a trend combo that combines inflation and economic activity. Column three in Table 1 reports the performance of this strategy, where the achieved diversification benefits results in a Sharpe ratio of 0.70. The trend combo inherits a negative skewness from the inflation variable while displaying moderate kurtosis. The cumulative return of this portfolio is also plotted in Figure 2. Analyzing the strategy’s worst and best months, we find the three worst months to be October 2008 (−4.46%), September 2008 (−4.25%), and July 1977 (−3.42%) and the three best months to be June 2003 (2.91%), December 2001 (2.91%), and January 2003 (3.06%). The last three columns of Table 1 report results on how the performance relates to the lookback period. We construct three combos with short-, medium-, and long-term trends, measured over 1–12 months, 13–36 months, and 37–60 months, respectively. The investment performance improves with the length of the trend used. The short-term trend portfolio has an annualized Sharpe ratio of 0.43 versus 0.80 for the long-term trend portfolio. Hence, the strongest return contribution to the trend strategy seems to come from long-term trends in fundamentals. That the trend strategies generate positive returns indicates that the strategies capture cross-sectional predictability of currency returns.

We run predictive panel regressions using the next period’s currency return as the dependent variable and standardized portfolio weights as independent variables:

$$R_{c,t+1} = a_t + b \tilde{w}_{c,t} + \epsilon_{c,t+1}, \quad (4)$$

where $R_{c,t+1}$ denotes the monthly excess return for currency c at time $t + 1$, $\tilde{w}_{c,t}$ denotes the standardized portfolio weight for currency c at time t , and a_t denotes time fixed effects. All panel regressions report standard errors clustered by currency. Clustering by currency and time makes little difference to the main results. Portfolio weights for aggregate strategies are constructed by weighting each sub-strategy by the inverse of its past volatility and scaling the weights such that they sum to one. The weights in the regressions are for the aggregate portfolios but cross-sectionally standardized in each month. Note that using weights as predictors is effectively the same as using the cross-sectional ranks as predictors. We predict returns using portfolio weights for trend strategies based on economic activity, inflation, and the trend combo. Table 2 reports results indicating that all trend strategies predict future currency returns with coefficients being highly statistically significant. For example, a coefficient of 0.13 for the trend combo means that a portfolio weight one standard deviation above the mean predicts a positive currency return of 0.13% the next month. An increase in the portfolio weight can also be interpreted in terms of a currency’s portfolio rank. In this case, a one-standard-deviation-increase in portfolio weight corresponds approximately to a four-step increase in ranking. There are on average 13 currencies in the portfolio at each time t , so, for example, moving from rank 3 to rank 11 predicts a 0.26% higher return the next month. The last column indicates that both trend strategies predict returns significantly when considered jointly. Hence, there seems to be complementary information in economic activity and inflation.

4.2 Economic momentum versus benchmark strategies

Having established that investing according to past trends in fundamentals yields high risk-adjusted returns, the question arises as to whether the economic momentum strategy is simply a repackaged version of standard currency strategies. We answer this question by relating the returns on trend strategies to those on carry, momentum, and value portfolios. We

construct the benchmark strategies as described in Section 3.3 and report their performance statistics in Table 3. The carry, momentum, and value strategies exhibit Sharpe ratios of 0.57, 0.25, and 0.42, respectively. Both carry and momentum have negative skewness while value displays positive skewness.

Figure 3 plots the cumulative returns of the three benchmark strategies. Several interesting observations are worth discussing. First, it is evident that the carry trade produces the largest cumulative return, albeit less than the trend combo, as was plotted in Figure 2. Second, the cumulative returns of momentum and value move in opposite directions over extended periods of time, indicating diversification benefits of combining the two strategies. We measure their return correlation to be -0.44 , as reported in the Online Appendix. Third, it is noteworthy that the returns of currency momentum seem to have flattened out post the financial crisis. Fourth, the value strategy performed strongly during the financial crisis while the carry trade suffered a sharp loss.

Next we study how well the benchmark strategies can explain the returns of the trend combo. We run contemporaneous time-series regressions using trend-combo returns as the dependent variable and the returns of various benchmark strategies as independent variables. Table 4 reports the results of each regression specification. First, we regress the returns of the trend combo on the carry, momentum, and value strategies. The resulting alpha is highly significant, both statistically and economically. The annualized alpha of 1.36% represents 58% of the average return of the trend combo (i.e., less than half of the average return of the trend combo is explained by the benchmarks). The positive and significant coefficient for carry returns suggests that a portion of the trend returns reflects cross-sectional differences in interest rate differentials. The coefficient for momentum is close to zero and insignificant, indicating that price momentum and economic momentum are unrelated. Second, we regress combo returns on proxies for volatility and funding risks in the forms of changes

in FX volatility, the VIX index, and the TED spread.⁹ All coefficients are negative and statistically significant. This suggests that the trend combo performs poorly in times of heightened volatility and funding risk.¹⁰ Noting that these proxies are not traded portfolios, we construct factor-mimicking portfolios by regressing the non-traded factors on five carry-sorted portfolios as in [Menkhoff, Sarno, Schmeling, and Schrimpf \(2012a\)](#). Regressing combo returns on the factor-mimicking portfolios still yields a large and strongly significant alpha of 1.37%. Hence, the trend combo generates returns over and above those of common currency strategies and risk factors.

Finally, we turn to predictive regressions. We begin by predicting currency returns one month ahead using portfolio weights for the carry, momentum, and value strategies as well as the trend combo. Portfolio weights are standardized in each month as described earlier. Specifications I–III in [Table 5](#) indicate that the three benchmark strategies positively predict returns with highly significant coefficients. However, the magnitudes of the coefficients and the R-squared values are smaller than those of the trend combo in specification IV. Specification V includes all benchmark strategies jointly, with coefficients remaining significant. Specification VI includes the trend combo in addition to the benchmark strategies. The combo enters with a strongly statistically significant coefficient while reducing the coefficients for carry and momentum. Splitting the sample period in half, we find the trend combo to have significant forecasting power of currency returns in both periods while the predictive ability of carry and momentum has been statistically insignificant during the second half of the sample period. See the Online Appendix for these results.

We also predict multi-period currency returns for up to 12 months. The regressions include the same three benchmark strategies as used for monthly returns together with

⁹We construct FX volatility as in [Menkhoff, Sarno, Schmeling, and Schrimpf \(2012a\)](#) and the TED spread is the three-month USD Libor minus the three-month US T-bill rate, where both rates are collected from *Datastream*.

¹⁰Relatedly, [Bakshi and Panayotov \(2013\)](#) find that measures of currency volatility and aggregate liquidity predict carry trade returns.

the trend combo. Table 6 reports the results. The coefficients of the trend combo remain high and strongly statistically significant for all forecasting horizons. In contrast, the carry and momentum strategies only have statistically significant forecasting power over the next month. Like the trend combo, the value strategy possesses significant forecasting power for all forecasting horizons. The R-squared increases from 0.84% for the one-month horizon to 11.01% for the 12-month horizon. Splitting the sample in half, the trend combo strongly predicts multi-period returns in both sample periods, as reported in the Online Appendix.

The Online Appendix also reports results from cross-sectional predictability regressions in terms of foreign currency appreciation or dollar depreciation, i.e., $(S_{c,t+1} - S_{c,t})/S_{c,t}$. Results from full-sample regressions suggest that the trend combo has no significant predictive power while carry predicts foreign currency appreciation with a negative coefficient over long horizons. That is, countries with relatively high (low) interest rates tend to experience a depreciating (appreciating) currency over the long term. In contrast, value predicts foreign currency appreciation with positive signs for all horizons. The predictive coefficients for momentum are statistically insignificant. When estimating the same regressions for the first and second half of the sample period we find that carry and the trend combo predict currency appreciations with negative signs over the first period but the predictive sign of the trend combo turns positive and statistically significant for the second half of the period. That is, the trend combo predicts currency appreciation positively starting in the 1990s. This suggests that economic activity and inflation, in the cross-section, started to predict changes in exchange rates positively during the 1990s. This is akin to [Molodtsova and Papell \(2009\)](#) who find that Taylor-rule fundamentals, in the time-series, started to predict changes in exchange rates positively in the early 1990s, when central banks are presumed to have started adopting some form of a Taylor rule. Our results suggest that this empirical finding is both a cross-sectional and a time-series phenomenon and that the conduct of monetary policy might be a potential explanation for our overall findings.

4.3 Economic momentum and the carry trade

The last sub-section demonstrated that the trend combo has a positive loading on the carry strategy when estimating time-series regressions, while still delivering a positive alpha. In addition, introducing the trend combo in a predictability regression with carry reduces the carry coefficient while even making it statistically insignificant in the second half of the sample period. We further investigate the relation between the carry strategy and the economic momentum strategy by regressing carry-trade returns on the returns of the remaining benchmark strategies and the various trend strategies. Table 7 reports the results. Specification I reports the results of including only the additional benchmark strategies as independent variables. The constant is statistically significant and represents an annualized alpha of 2.83%. Hence, existing currency strategies have problems explaining carry-trade returns, as is well established in the literature. Specifications II–IV include trend strategies consisting of economic activity, inflation, and the combo. We find that all trend strategies have positive and strongly significant coefficients. Inflation and the combo reduce the magnitude of the carry alpha to 0.62% and 0.99%, respectively, and render the alpha of carry trades statistically insignificant. Specifications V–VII report results from regressing carry returns on combo returns of different lookback periods, where short, medium, and long-term trends refer to 1–12, 13–36, and 37–60 months, respectively. Results show that economic momentum over the medium and long term subsume the carry alpha.

While our results suggest that the economic momentum strategy subsumes the alpha of carry trades one might wonder how portfolio weights for the two strategies differ. We construct two intuitive measures. The first measure is the fraction of months each currency weight is below or above the monthly median weight for the particular strategy. Table 8 shows a significant variation across strategies and countries. The three countries whose currencies differ most in their distribution between the two strategies are France, Israel, and New Zealand. The second measure is the fraction of months the combo and carry weights for

a particular currency have different signs (long versus short positions). Table 8 shows that four currencies display different signs more than half of the time while 14 currencies have different signs one third of the time or more. On average, across currencies, the weights have different signs 38% of the time. We also compare the weights of the economic momentum strategy to those of the momentum and value strategies and find them to have different signs 44% and 52% of the time, respectively, as reported in the Online Appendix. Summing up, the analysis suggests that combo weights and benchmark weights display different dynamics.

4.4 Economic momentum in a diversified portfolio

The importance of the economic momentum strategy can also be gauged from a portfolio investment perspective. Table 9 reports the weights in the ex post mean-variance efficient portfolio for combinations of carry, momentum, and value benchmarks, and the trend combo. The resulting portfolio Sharpe ratios are also reported. Columns I–IV report portfolios based on individual strategies with similar levels and differences in Sharpe ratios as before. Column V reports the ex post mean-variance efficient portfolio of the three benchmarks. The optimal allocation would have been 37% in carry, 23% in momentum, and 40% in value. The Sharpe ratio would have been 0.78, considerably higher than the Sharpe ratios of the benchmarks individually. This confirms the diversification benefits shown and discussed in Jordà and Taylor (2012), Asness, Moskowitz, and Pedersen (2013), Kroencke, Schindler, and Schrimpf (2014), and Barroso and Santa-Clara (2015a). Interestingly, when the economic momentum strategy is added to the portfolio in column VI, its weight is 45%. That is, the economic momentum strategy receives the largest allocation of the ex post mean-variance efficient portfolio. The weights in momentum and value are reduced by approximately one third and the weight in carry drops by two thirds to 12% while turning statistically insignificant. The portfolio Sharpe ratio increases to 0.91. Again, this highlights the subsumption of carry by economic momentum.

4.5 Volatility timing

We have considered a further refinement when constructing the portfolios. We scaled the final weights of a portfolio by a factor of 5% divided by the past volatility of the strategy. This represents a simple method of volatility timing ([Fleming, Kirby, and Ostdiek, 2001](#); [Kirby and Ostdiek, 2012](#)) and leads to improvements for momentum strategies in equity markets ([Barroso and Santa-Clara, 2015b](#); [Daniel and Moskowitz, 2016](#)) and carry trades in currency markets ([Daniel, Hodrick, and Lu, 2017](#)). More specifically, volatility timing has shown to give higher Sharpe ratios, and less skewed and leptokurtic returns. We found modest improvements in Sharpe ratios for the trend combo while the performance of the economic-activity combo dropped somewhat and the performance of the inflation combo improved considerably. Moreover, we found minor differences in skewness but a significant drop in kurtosis when employing volatility timing. That volatility timing somewhat increases Sharpe ratios while leading to a trend combo with only moderate values in skewness and kurtosis suggests that the performance of the trend combo is unrelated to skewness or kurtosis risks.

In the Online Appendix we report regressions of the returns with volatility timing on the returns without volatility timing. The inflation combo with volatility timing yields a statistically and economically significant alpha, whereas the economic activity combo with volatility timing shows an insignificant alpha. Among the benchmark strategies, momentum shows a significant volatility timing alpha whereas the carry and value alphas are not significant. Our results are overall in line with [Moreira and Muir \(2017\)](#), who find large and positive alphas for several equity factors but to a less extent so for the carry trade. Our finding that volatility timing matters is also in line with [Della Corte, Sarno, and Tsiakas \(2009\)](#), who study the predictive ability of forward premia and find that volatility timing yields significant economic value to an investor. Importantly, while volatility timing improves the absolute performance, the relative performance between the economic momentum strategies

and benchmarks is not changed that much.

4.6 Transaction costs

It is reasonable to ask whether the profits of the trend strategies survive when taking transaction costs into account. Regarding which level of transaction costs to use, [Menkhoff, Sarno, Schmeling, and Schrimpf \(2012a\)](#) report in their working paper version an average bid-ask spread for spot and forward rates of approximately ten basis points over the 1976–2010 period for developed markets. [Mancini, Ranaldo, and Wrampelmeyer \(2013\)](#) report average bid-ask spreads of approximately five basis points for the most liquid currency pairs over the 2007–2009 period, albeit with occasional spikes during the financial crisis. An investor incurs a cost of half the spread when buying or selling a currency. We have considered various assumptions on the bid-ask spread. The Sharpe ratio of the combo goes from 0.70 to 0.64 at a ten basis points spread.

While the combo is still profitable after assuming a constant bid-ask spread, a more elaborate cost analysis would need to consider the significant time and cross-sectional variation in foreign exchange liquidity as shown by [Karnaukh, Ranaldo, and Söderlind \(2015\)](#). For example, it is likely that the various strategies would experience larger transaction costs in times of increased volatility and funding constraints and in periods when liquidity simultaneously deteriorates in bond and equity markets. Relatedly, the transaction costs in emerging markets would be much higher than for developed markets. For example, assuming a bid-ask spread of thirty basis points lowers the Sharpe ratio of the emerging market combo from 0.76 to 0.66.

An additional feature that affects transaction costs is whether volatility timing (as discussed above) is employed. Without volatility timing, changes in portfolio weights and therefore transaction costs are functions of the variation in investment signals. With volatility timing, transaction costs are also functions of time-variation in portfolio volatility. Hence,

the value of volatility timing is ultimately a tradeoff between increased costs and turnover and any potential investment gains. See [Baltas and Kosowski \(2017\)](#) for a detailed discussion of the impact of volatility estimators on portfolio turnover and costs.

To sum up, the combo remains profitable after assuming a constant bid-ask spread and its high relative performance to the benchmarks remains. Hence, investing in currencies based on past trends in fundamentals seems profitable also after transaction costs.

4.7 G10 and emerging data, sub-samples, and alternative portfolios weights

We make several robustness checks. First, we evaluate the performance of the economic momentum strategy on data for the G10 and emerging market countries. Overall the results are similar. The combo displays a lower Sharpe ratio when using the G10 data but a higher Sharpe ratio in emerging markets (0.55 and 0.76, respectively, versus 0.70 for developed markets). Both G10 and emerging market data result in equal or less negatively skewed returns (-0.61 and -0.12 , respectively, versus -0.65) and less excess kurtosis (2.48 and 2.39, respectively, versus 2.92). The Online Appendix reports further details on the performance for the G10 and emerging market countries.

Second, we split our sample in half (1976–1996 and 1996–2017) and consider the performance of the trend strategies and the predictability of currency returns in the two sub-samples. As mentioned earlier, the availability of macro data varies over time and countries and this is particularly true for the first half of the sample period. We find that the trend combo delivered similar performance and return properties in both sub-periods with a Sharpe ratio of 0.73 and a skewness of -0.71 in the first sub-sample and with a Sharpe ratio of 0.69 and a skewness of -0.63 in the second sub-sample. We also run predictive panel regressions for the two sub-samples and find that the trend combo significantly predicts excess returns

in both, controlling for the carry, momentum, and value strategies. The Online Appendix reports more detailed results.

Third, while we report results on ranked-based weighting, we have considered different ways of constructing the portfolios. For example, a common approach in the literature is to sort currencies into percentiles based on signals, and then form long-short portfolios. We find similar results when the benchmarks and trend combo are constructed as long-short portfolios based on quintiles rather than with ranked-based weighting.

4.8 Time-series strategies

While our main results consider trend and benchmark strategies that are dollar neutral, we also consider strategies with dollar exposure. We construct trend strategies with portfolio weights based on the sign of a signal:

$$w_{c,i,l,t} = \text{sign}(z_{c,i,l,t} - z_{US,i,l,t})/C_t, \quad (5)$$

where $z_{c,i,l,t}$ is defined as in Section 3.1 and denotes the trend signal for currency c 's fundamentals for fundamental index i over the lookback period l at time t , $z_{US,i,l,t}$ denotes the corresponding signal for US fundamentals, and C_t denotes the number of available currencies. We denote this strategy time-series economic momentum, measuring the difference in fundamental trends between a foreign country and the home country (here US). The strategy goes long (short) currency c when its country exhibits stronger (weaker) economic momentum than the US. Consequently, this strategy can be either long or short the dollar.

We find that the time-series combo delivers a somewhat lower Sharpe ratio compared to the cross-sectional combo, 0.59 versus 0.70. Furthermore, the time-series combo has a skewness of returns close to zero but a higher kurtosis than the cross-sectional combo. While the cross-sectional combo displays a Sharpe ratio close to 0.70 for both the first and second

half of the sample period the time-series combo exhibits a Sharpe ratio close to zero during the first half and a Sharpe ratio of 1.14 during the second half. The difference in performance is partly explained by the fact that the time-series combo started to predict spot returns positively in the 1990s. This is the same time as Taylor-rule fundamentals start to predict spot returns in the time-series, as we discussed in Section 4.2. The Online Appendix reports summary statistics, plots cumulative returns of the time-series trend strategies, and provides correlations between cross-sectional and time-series trend strategies.

We construct time-series signals also for the benchmark strategies. Portfolio weights for the time-series benchmark strategies are defined as $w_{c,t} = \text{sign}(\text{signal}_{c,t})/C_t$, where $\text{signal}_{c,t}$ denotes one of the benchmark signals in Section 3.3. More specifically, the carry strategy is based on the forward premium (or interest rate differential), the momentum strategy on the past 12-month return while the value strategy reflects the negative of the log change in the real exchange rate over the past 60 months. Note that the benchmark signals are already, effectively, measured against the base currency USD. The literature has considered similar time-series strategies; for example, [Daniel, Hodrick, and Lu \(2017\)](#) consider the forward premium signal, and [Moskowitz, Ooi, and Pedersen \(2012\)](#) and [Baltas and Kosowski \(2017\)](#) consider the momentum signal. We find that the time-series versions of carry and momentum have higher Sharpe ratios than the cross-sectional versions while the time-series value strategy has a negative Sharpe ratio. In addition, we construct the dollar-carry factor that [Lustig, Roussanov, and Verdelhan \(2014\)](#) study. The Online Appendix reports summary statistics on the time-series benchmark strategies and provide plots of their cumulative returns.

We check if the returns of the time-series combo can be explained by the time-series benchmark strategies and factors linked to volatility and funding risks. The Online Appendix reports results on these regressions. We find that the returns of the time-series combo cannot be explained by the time-series benchmark strategies, delivering a statistically signif-

icant alpha of 1.03% which represents 72% of the average return on the time-series combo. Hence, only 28% of the returns of the time-series combo can be attributed to the benchmark strategies. Similar to the cross-sectional combo, the time-series combo has negative loadings on volatility and funding risks. Still, our findings suggest that both the cross-sectional and time-series combos remain unexplained by common currency strategies.

Finally, we analyze how the cross-sectional and time-series combos relate to each other. We follow [Moskowitz, Ooi, and Pedersen \(2012\)](#) and [Goyal and Jegadeesh \(2018\)](#), and regress the returns of the cross-sectional combo on the time-series combo and vice versa. We consider the full sample period, the first half of the sample (1976–1996), and the second half of the sample (1996–2017). We find that the cross-sectional combo has a statistically and economically significant alpha with respect to the time-series combo of 1.39% which represents 59% of the average return of the cross-sectional combo. We also find that the time-series combo has an economically significant alpha with respect to the cross-sectional combo of 43% of its average return, albeit not statistically significant. Interestingly, we find significant differences between the first and second sub-samples. While the cross-sectional combo exhibits a large and statistically significant alpha with respect to the time-series combo during the first half of the sample, the opposite is true for the second half. The large and significant alpha of the time-series combo from 1996 and onwards is consistent with its superior Sharpe ratio over that period.

5 Conclusion

We find that past trends in fundamentals positively predict currency returns. A strategy that goes long (short) currencies whose countries have experienced strong (weak) economic momentum returns a Sharpe ratio of 0.70 and yields a significant alpha when controlling for standard currency strategies. Returns on the economic momentum strategy subsume the

alpha of carry trades, which suggests that sorting on carry is captured by sorting on past trends in fundamentals. That more than half of the economic momentum returns are left unexplained by standard strategies suggests that investing according to economic momentum represents an additional source of returns. We discuss possible explanations for the returns, but their exact source remains an open question. More generally, our findings suggest it might be fruitful to analyze other asset classes and the momentum in their respective return drivers. We leave such interesting research for the future.

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Table 1: Performance of aggregated trend strategies and the trend combo for different lookback periods

	Trend variables			Lookback period for the combo		
	Economic activity	Inflation	Combo	Short term	Medium term	Long term
Mean	1.522	3.279	2.346	1.310	2.341	2.930
Standard deviation	3.905	4.971	3.328	3.038	3.704	3.658
Skewness	0.275	-0.778	-0.655	-0.610	-0.610	-0.518
Excess kurtosis	1.823	5.451	2.916	2.432	3.545	1.810
AR(1)	-0.040	0.062	0.014	-0.053	0.015	0.040
Sharpe ratio	0.390	0.660	0.705	0.431	0.632	0.801

The table presents performance measures for sub-combos (economic activity and inflation) and the diversified trend combo using different lookback periods. Economic activity combines industrial production, retail sales, and the inverse of unemployment; inflation combines consumer and producer price indices. Short-term trends use lookback periods of 1–12 months, medium-term trends use 13–36 months, and long-term trends use 37–60 months. The strategies use rank-based weights each month, going long currencies with relatively strong trends and shorting currencies with relatively weak trends. Each strategy diversifies across sub-strategies by weighting each sub-strategy by the inverse of its past volatility. The measures are based on monthly returns, but means, standard deviations, and Sharpe ratios are annualized. AR(1) refers to the first-order autocorrelation of returns. The sample period is January 1976 to March 2017.

Table 2: Using trend strategies to predict currency returns

	Currency returns			
	I	II	III	IV
Economic activity	0.066 (0.019)			0.061 (0.020)
Inflation		0.134 (0.022)		0.132 (0.021)
Trend combo			0.130 (0.020)	
Adjusted R^2 (%)	0.51	0.11	0.48	0.60
Number of observations	6472	6472	6472	6472

The table presents the results of panel regressions in which next month's currency returns are predicted using the current month's portfolio weights: $R_{c,t+1} = a_t + b\tilde{w}_{c,t} + \epsilon_{c,t+1}$, where $R_{c,t+1}$ denotes the monthly excess return for currency c at time $t+1$ and $\tilde{w}_{c,t}$ denotes the standardized portfolio weight for currency c at time t . Portfolio weights are constructed by weighting all sub-strategies by the inverses of their respective past volatilities, scaling the weights such that they sum to one. The weights in the regressions are for the aggregate portfolios but cross-sectionally standardized in each month. Weights for economic activity are combined weights of industrial production, retail sales, and the inverse of unemployment; weights for inflation are combined weights for consumer and producer prices. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to March 2017.

Table 3: Performance of benchmark strategies

	Carry	Momentum	Value
Mean	3.094	1.606	2.578
Standard deviation	5.429	6.333	6.146
Skewness	-0.283	-0.424	0.423
Excess kurtosis	1.481	3.152	2.076
AR(1)	-0.005	0.019	0.074
Sharpe ratio	0.569	0.254	0.420

The table presents performance measures for carry, momentum, and value strategies. Each month, a cross-sectional strategy weights currencies according to their carry, momentum, or value rank. The measures are based on monthly returns, but means, standard deviations, and Sharpe ratios are annualized. AR(1) refers to the first-order autocorrelation of returns. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

Table 4: Returns of the trend combo on benchmark strategies and factors

	Returns on trend combo		
	I	II	III
Constant	1.356 (0.455)	2.382 (0.642)	1.370 (0.466)
Carry	0.300 (0.046)		
Momentum	-0.008 (0.049)		
Value	-0.007 (0.063)		
Δ FXVOL		-0.012 (0.005)	
Δ VIX		-0.057 (0.021)	
Δ TED		-0.005 (0.002)	
FMP FXVOL			-0.185 (0.051)
FMP VIX			0.002 (0.001)
FMP TED			-0.086 (0.020)
Adjusted R^2 (%)	24.74	11.45	25.00
Number of observations	434	326	490

The table presents a contemporaneous regression of the monthly returns of the trend combo on benchmark strategies and changes in measures of volatility and funding conditions. Specification I uses the carry, momentum, and value benchmarks. Specification II uses changes in foreign exchange volatility (FXVOL), the VIX volatility index, and the TED spread, which are all non-traded factors. Specification III uses factor mimicking portfolios (FMPs), which are constructed from regressions of the non-traded factors on five carry-sorted portfolios as in [Menkhoff, Sarno, Schmeling, and Schrimpf \(2012a\)](#) for the longest available sample of factors. The factors are then mimicked on longest sample period possible. Constants in specifications I and III are annualized alphas, whereas the constant in specification II cannot be interpreted as alpha as the factors are non-traded. Point estimates are reported with [Newey and West \(1987\)](#) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. Adjusted R^2 values and the number of observations are also reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods. Specification II starts in January 1990 due to data availability for the VIX volatility index.

Table 5: Using benchmark strategies and the trend combo to predict currency returns

	Currency returns					
	I	II	III	IV	V	VI
Carry	0.117 (0.017)				0.086 (0.028)	0.057 (0.028)
Momentum		0.074 (0.027)			0.071 (0.037)	0.058 (0.036)
Value			0.096 (0.026)		0.109 (0.034)	0.110 (0.029)
Trend combo				0.130 (0.020)		0.093 (0.019)
Adjusted R^2 (%)	0.38	0.14	0.25	0.48	0.63	0.84
Number of observations	6532	6280	5541	6472	5493	5493

The table presents the results of panel regressions in which next month's currency returns are predicted using the current month's portfolio weights: $R_{c,t+1} = a_t + b\tilde{w}_{c,t} + \epsilon_{c,t+1}$, where $R_{c,t+1}$ denotes the monthly excess return for currency c at time $t + 1$ and $\tilde{w}_{c,t}$ denotes the standardized portfolio weight of currency c at time t . Portfolio weights of the trend combo are constructed by weighting all sub-strategies by the inverses of their respective past volatilities, scaling the weights such that they sum to one. The weights in the regressions are for the aggregate portfolios but cross-sectionally standardized in each month. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

Table 6: Predicting currency returns over various forecasting horizons

	Cumulative currency returns			
	1 month	3 months	6 months	12 months
Carry	0.057 (0.028)	0.024 (0.029)	0.020 (0.032)	0.015 (0.037)
Momentum	0.058 (0.036)	0.026 (0.022)	0.007 (0.014)	-0.018 (0.016)
Value	0.110 (0.029)	0.125 (0.028)	0.123 (0.031)	0.111 (0.031)
Trend combo	0.093 (0.019)	0.116 (0.017)	0.123 (0.019)	0.141 (0.022)
Adjusted R^2 (%)	0.84	2.61	5.14	11.01
Number of observations	5493	5430	5367	5241

The table presents the results of panel regressions in which future cumulative currency returns over one-, three-, six-, and 12-month horizons are predicted along with current standardized portfolio weights of three benchmark strategies (i.e., carry, momentum, and value) and the trend combo. The cumulative returns are scaled to a monthly basis to facilitate the comparison of the coefficients. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

Table 7: Using trend strategies to explain carry trade returns

	I	II	III	IV	V	VI	VII
Constant	2.832 (0.943)	2.488 (0.929)	0.618 (0.767)	0.995 (0.797)	2.040 (0.861)	1.245 (0.785)	0.768 (0.840)
Momentum	0.127 (0.085)	0.113 (0.083)	0.138 (0.053)	0.102 (0.061)	0.146 (0.069)	0.095 (0.060)	0.089 (0.063)
Value	0.007 (0.097)	0.002 (0.090)	0.034 (0.054)	0.011 (0.062)	-0.011 (0.070)	-0.001 (0.061)	0.032 (0.065)
Economic activity		0.288 (0.115)					
Inflation			0.660 (0.065)				
Trend combo				0.833 (0.110)			
Trend combo (short term)					0.701 (0.137)		
Trend combo (medium term)						0.746 (0.101)	
Trend combo (long term)							0.721 (0.094)
Adjusted R^2 (%)	1.56	5.58	35.02	25.97	15.12	26.22	24.29
Number of observations	434	434	434	434	434	434	434

The table presents contemporaneous regressions of carry-trade returns on benchmark strategies and the trend strategies. Point estimates are reported with [Newey and West \(1987\)](#) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. Adjusted R^2 values and the number of observations are also reported. The constants are alphas and expressed in % per year. The sample period is January 1976 to March 2017 but returns start at different dates due to differences in lookback periods.

Table 8: Weight analysis for the carry strategy and the trend combo

	Carry		Trend combo		Different signs
	Below median	Above median	Below median	Above median	
Australia	0.099	0.901	0.131	0.869	0.181
Austria	0.929	0.071	0.622	0.378	0.424
Belgium	0.690	0.310	0.696	0.304	0.476
Canada	0.351	0.649	0.462	0.538	0.472
Denmark	0.380	0.620	0.474	0.526	0.517
Eurozone	0.705	0.295	0.727	0.273	0.316
Finland	1.000	0.000	0.750	0.250	0.250
France	0.338	0.662	0.801	0.199	0.554
Germany	0.825	0.175	0.919	0.081	0.104
Ireland	0.211	0.789	0.304	0.696	0.456
Israel	0.564	0.436	0.038	0.962	0.590
Italy	0.610	0.390	0.236	0.764	0.522
Japan	0.681	0.319	0.865	0.135	0.374
Netherlands	0.771	0.229	0.867	0.133	0.292
New Zealand	0.046	0.954	0.541	0.459	0.481
Norway	0.368	0.632	0.336	0.664	0.408
Portugal	0.298	0.702	0.026	0.974	0.299
Spain	0.447	0.553	0.059	0.941	0.437
Sweden	0.499	0.501	0.559	0.441	0.405
Switzerland	0.899	0.101	0.802	0.198	0.165
United Kingdom	0.188	0.812	0.290	0.710	0.346
Average					0.384

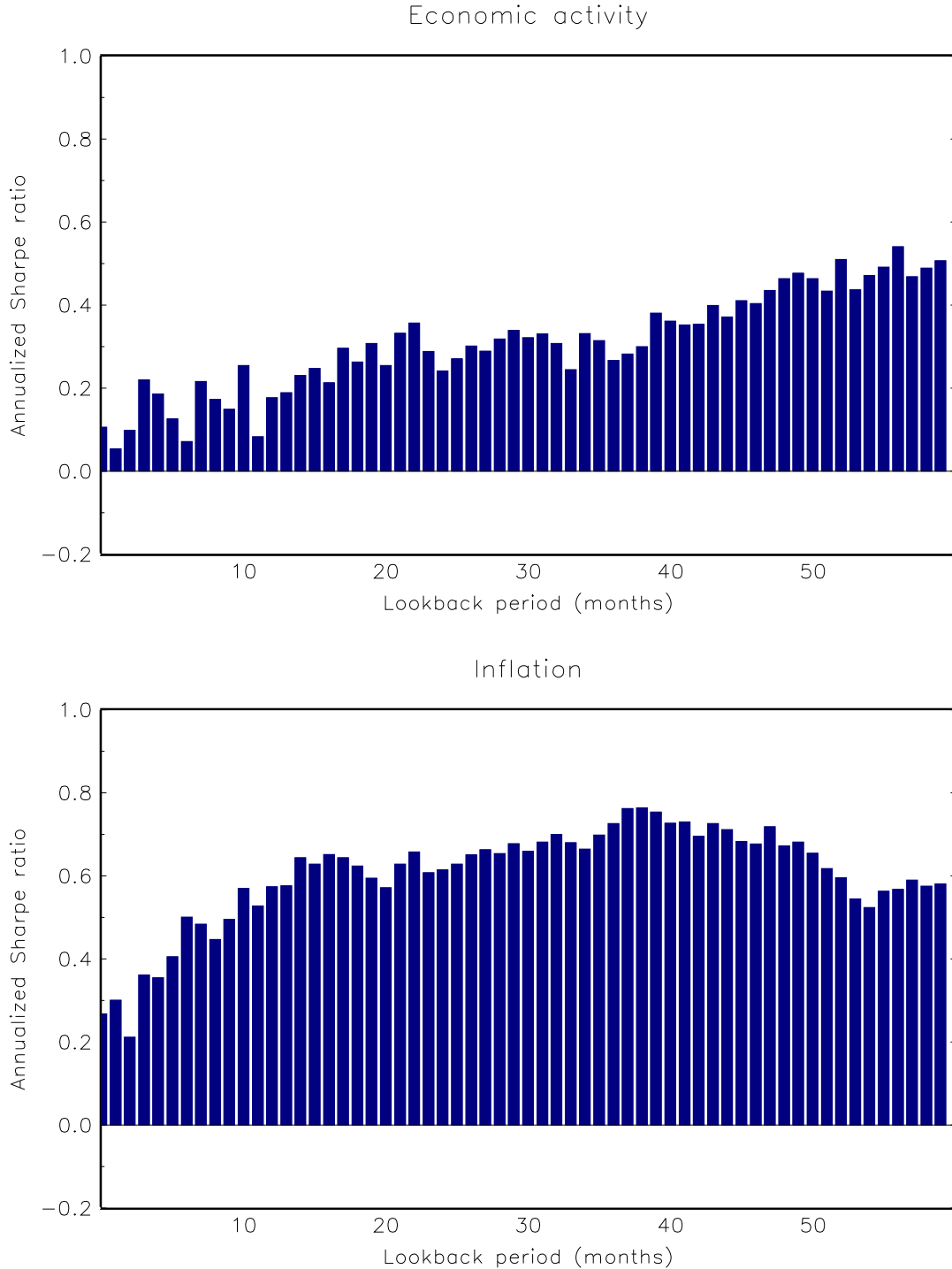
The table presents an analysis of the portfolio weights for the carry benchmark and the trend combo. The first two columns report the fraction of months each currency weight is below or above the monthly median weight for the carry strategy. By construction, the two columns sum to one. The next two columns report the corresponding fractions for the trend combo. The final column reports the fraction of months the carry and combo weights for a particular currency have different signs. The last row in that column reports the average fraction across the currencies. The sample period is January 1976 to March 2017.

Table 9: Weights in ex post mean-variance efficient portfolios

	I	II	III	IV	V	VI
Carry	1.000				0.372 (0.111)	0.118 (0.096)
Momentum		1.000			0.232 (0.090)	0.162 (0.068)
Value			1.000		0.395 (0.075)	0.272 (0.070)
Trend combo				1.000		0.448 (0.123)
Sharpe ratio	0.540	0.176	0.420	0.676	0.779	0.913

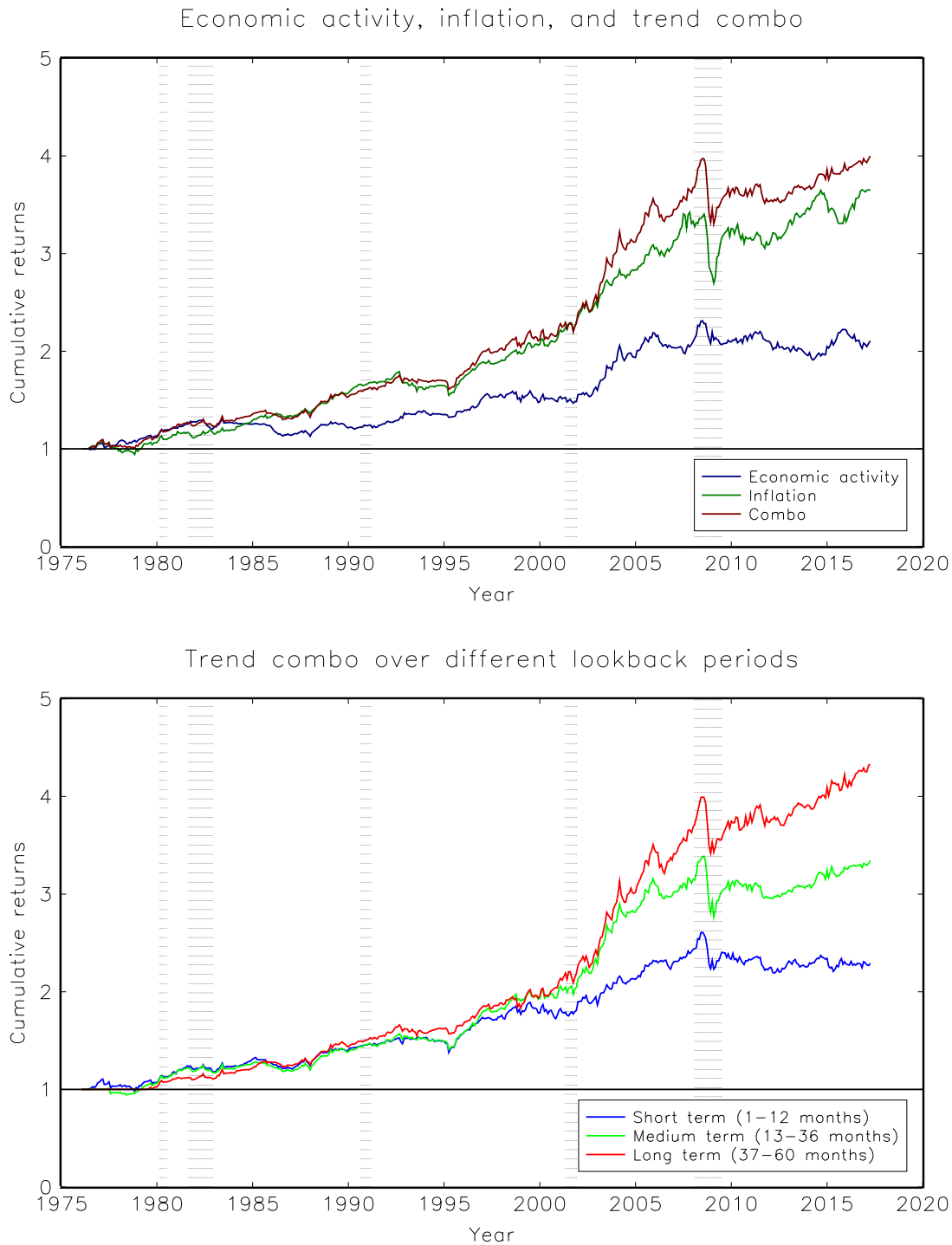
The table presents the weights in the ex post mean-variance efficient portfolio for various combinations of benchmarks (carry, momentum, and value) and the trend combo, and the resulting portfolio Sharpe ratio. When there are multiple strategies in the portfolio, weights are reported with [Newey and West \(1987\)](#) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. The portfolio Sharpe ratios are annualized. The sample period is January 1976 to March 2017, but due to the long lookback period for the value benchmark all returns start in February 1981.

Figure 1: Sharpe ratios of trend strategies



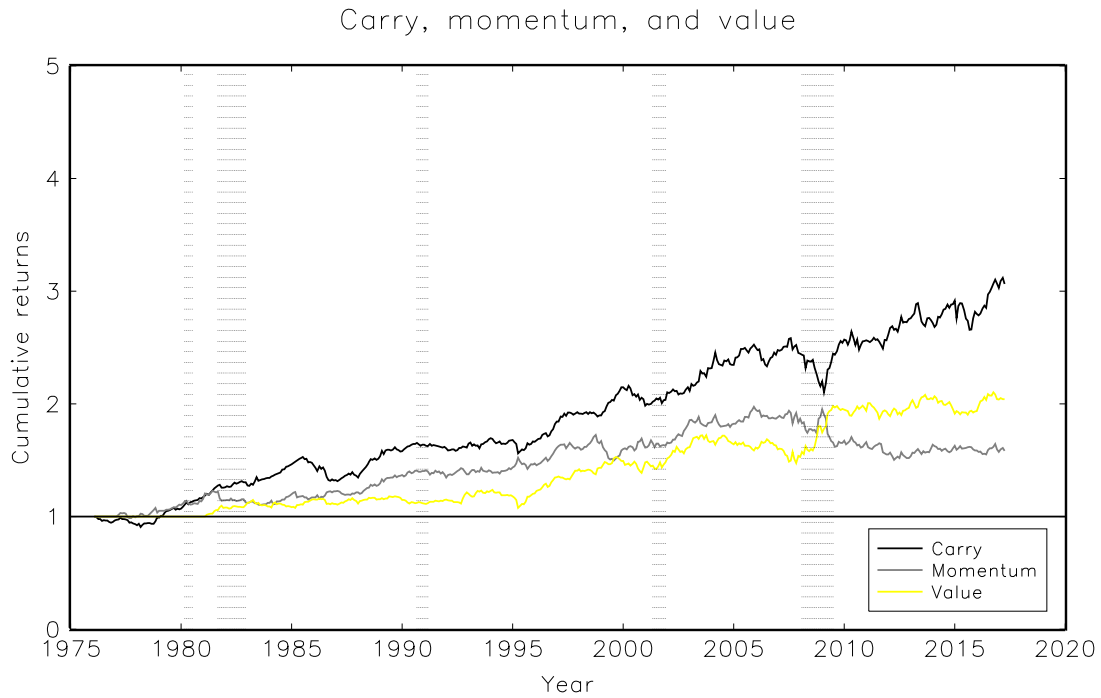
The figure shows annualized Sharpe ratios for the two sub-combos, economic activity and inflation, over a range of lookback periods. Economic activity combines industrial production, retail sales, and the inverse of unemployment; inflation combines consumer and producer price indices. The strategies use rank-based weights each month for a particular lookback period, going long currencies with relatively strong trends and shorting currencies with relatively weak trends. The sample period is January 1976 to March 2017 but returns start at different dates depending on the lookback period.

Figure 2: Cumulative portfolio returns of trend strategies



The figure shows the cumulative returns of trend strategies based on fundamental variables. All portfolio returns are scaled ex post as to achieve an annualized ex post volatility of 5%. The color scheme in the top figure is: economic activity (blue), inflation (green), and trend combo (red); the color scheme in the bottom figure is: short-term (light blue), medium-term (light green), and long-term (light red) lookback periods. The shaded areas indicate US contractions (peak to trough) as dated by the NBER. The sample period is January 1976 to March 2017 but cumulative returns start at different dates due to differences in lookback periods.

Figure 3: Cumulative portfolio returns of benchmark strategies



The figure shows the cumulative returns of the benchmark strategies carry, momentum, and value. All portfolio returns are scaled ex post as to achieve an annualized ex post volatility of 5%. The color scheme is: carry (black), momentum (gray), and value (yellow). The shaded areas indicate US contractions (peak to trough) as dated by the NBER. The sample period is January 1976 to March 2017 but cumulative returns start at different dates due to differences in lookback periods.

Online Appendix for
“Economic Momentum and Currency Returns”

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1 Data cleaning description

There are outliers in data on 10/16/2003, when forward exchange rates against the GBP for Canada, Denmark, Japan, Norway, Sweden, and Switzerland more than double in value. We replace these observations by their previous trading day's values. Moreover, we amend the data as follows:

- China: The currency was pegged to USD up to 7/21/2005. We start Chinese data thereafter.
- Indonesia: Forward exchange rates against the GBP have a long series of repeated values between 2/16/2001 and 1/13/2003. We start after that period, on 1/14/2003. Similarly, forward exchange rates against the USD have a long series of repeated values between 2/16/2001 and 6/1/2007. We start after that period, on 6/4/2007.
- Ireland: Forward exchange rates and spot rates against the GBP have long streaks of 1s at the start of the sample period. We replace these with missing values and start the series on 3/30/1979.
- Lithuania: The consumer price index has an initial observation in January 1990 and then a series of missing values. We start the series after those missing values in December 1990.
- Turkey: Forward exchange rates against the GBP have a long series of repeated values between 2/20/2001 to 12/21/2001. We start after that period, on 12/24/2001.

A number of countries in the sample either joined the EUR or pegged their currency against the EUR during the sample period. These currencies are not active in the trend strategies after that particular date, as follows:

- 1/1/1999: Austria, Belgium, Finland, France, Germany, Ireland, Italy, Netherlands, Spain, and Portugal joined the EUR while Denmark pegged its currency to the EUR.
- 6/19/2000: The Greek currency was pegged against the EUR.
- 6/28/2004: The Slovenian currency was pegged against the EUR.
- 5/2/2005: The Latvian currency was pegged against the EUR.
- 11/28/2005: The Slovak currency was pegged against the EUR.

Appendix Table 1: Currency classifications and sample periods

	G10	Developed	Emerging	Begin	End
Australia	Yes	Yes		12/31/1996	3/31/2017
Austria		Yes		1/30/1976	12/31/1998
Belgium		Yes		1/30/1976	12/31/1998
Brazil			Yes	3/31/2004	3/31/2017
Canada	Yes	Yes		1/30/1976	3/31/2017
Chile			Yes	3/31/2004	3/31/2017
China			Yes	7/29/2005	3/31/2017
Colombia			Yes	3/31/2004	3/31/2017
Czech Republic			Yes	12/31/1996	3/31/2017
Denmark		Yes		1/30/1976	12/31/1998
Eurozone / Germany	Yes	Yes		1/30/1976	3/31/2017
Finland		Yes		12/31/1996	12/31/1998
France		Yes		1/30/1976	12/31/1998
Greece			Yes	12/31/1996	5/31/2000
Hungary			Yes	10/31/1997	3/31/2017
Iceland			Yes	3/31/2004	3/31/2017
India			Yes	10/31/1997	3/31/2017
Indonesia			Yes	1/31/2003	3/31/2017
Ireland		Yes		3/30/1979	12/31/1998
Israel		Yes		3/31/2004	3/31/2017
Italy		Yes		1/30/1976	12/31/1998
Japan	Yes	Yes		6/30/1978	3/31/2017
Latvia			Yes	3/31/2004	4/29/2005
Mexico			Yes	12/31/1996	3/31/2017
Netherlands		Yes		1/30/1976	12/31/1998
New Zealand	Yes	Yes		12/31/1996	3/31/2017
Norway	Yes	Yes		1/30/1976	3/31/2017
Poland			Yes	2/28/2002	3/31/2017
Portugal		Yes		1/30/1976	12/31/1998
Russia			Yes	3/31/2004	3/31/2017
Slovak Republic			Yes	2/28/2002	10/31/2005
South Africa			Yes	12/31/1996	3/31/2017
South Korea			Yes	2/28/2002	3/31/2017
Spain		Yes		1/30/1976	12/31/1998
Sweden	Yes	Yes		1/30/1976	3/31/2017
Switzerland	Yes	Yes		1/30/1976	3/31/2017
Turkey			Yes	12/31/2001	3/31/2017
United Kingdom	Yes	Yes		1/30/1976	3/31/2017
United States	Yes	Yes		1/30/1976	3/31/2017

The table presents the classification of countries into G10, developed markets, and emerging markets currencies. The classification for developed and emerging markets follows the MSCI classification. The beginning and ending dates reflect availability of financial and macro data to form the trend combo. Currencies now in the Eurozone are used in developed markets until December 31, 1998, after which the Euro is used. Before January 1, 1999, Germany is used in G10 and developed markets.

Appendix Table 2: *Datastream* mnemonics for spot and forward exchange rates

	Currency	Spot (GBP)	Forward (GBP)	Forward (USD)
Australia	AUD	AUSTDOL	UKAUD1F	USAUD1F
Austria	ATS	AUSTSCH	AUSTS1F	USATS1F
Belgium	BEF	BELGLUX	BELXF1F	USBEF1F
Brazil	BRL	BRACRUZ	UKBRL1F	USBRL1F
Canada	CAD	CNDOLLR	CNDOL1F	USCAD1F
Chile	CLP	CHILPES	UKCLP1F	USCLP1F
China	CNY	CHIYUAN	UKCNY1F	USCNY1F
Colombia	COP	COLUPES	UKCOP1F	USCOP1F
Czech Republic	CZK	CZECHCM	UKCZK1F	USCZK1F
Denmark	DKK	DANISHK	DANIS1F	USDKK1F
Eurozone	EUR	EURSTER	UKEUR1F	USEUR1F
Finland	FIM	FINMARK	UKFIM1F	USFIM1F
France	FRF	FRENFRA	FRENF1F	USFRF1F
Germany	DEM	DMARKER	DMARK1F	USDEM1F
Greece	GRD	GREDRAC	UKGRD1F	USGRD1F
Hungary	HUF	HUNFORT	UKHUF1F	USHUF1F
Iceland	ISK	ICEKRON	UKISK1F	USISK1F
India	INR	INDRUPE	UKINR1F	USINR1F
Indonesia	IDR	INDORUP	UKIDR1F	USIDR1F
Ireland	IEP	IPUNTER	IPUNT1F	USIEP1F
Israel	ILS	ISRSHEK	UKILS1F	USILS1F
Italy	ITL	ITALIRE	ITALY1F	USITL1F
Japan	JPY	JAPAYEN	JAPYN1F	USJPY1F
Latvia	LVL	LATVLAT	UKLVL1F	USLVL1F
Mexico	MXN	MEXPESO	UKMXN1F	USMXN1F
Netherlands	NLG	GUILDER	GUILD1F	USNLG1F
New Zealand	NZD	NZDOLLR	UKNZD1F	USNZD1F
Norway	NOK	NORKRON	NORKN1F	USNOK1F
Poland	PLN	POLZLOT	UKPLN1F	USPLN1F
Portugal	PTE	PORTESC	PORTS1F	USPTE1F
Russia	RUB	CISRUBM	UKRUB1F	USRUB1F
Slovak Republic	SKK	SLOVKOR	UKSKK1F	USSKK1F
South Africa	ZAR	COMRAND	UKZAR1F	USZAR1F
South Korea	KRW	KORSWON	UKKRW1F	USKRW1F
Spain	ESP	SPANPES	SPANP1F	USESP1F
Sweden	SEK	SWEKRON	SWEDK1F	USSEK1F
Switzerland	CHF	SWISSFR	SWISF1F	USCHF1F
Turkey	TRY	TURKLIR	UKTRY1F	USTRY1F
United Kingdom	GBP			USGBP1F
United States	USD	USDOLLR	USDOL1F	

The table presents the currency swift codes and *Datastream* mnemonics for spot exchange rates versus the GBP, and one-month forward exchange rates versus the GBP and the USD. We use forward rates against the British pound up to February 1, 2007, as they provide the longest available history, after which we use forward rates against the USD. The data are further checked and cleaned as described in the main text and the Online Appendix.

Appendix Table 3: Relationship between fundamentals and industrial production

	Retail sales	Unemployment	CPI	PPI
Industrial production	0.215 (0.031)	1.100 (0.116)	0.055 (0.037)	0.165 (0.041)
Adjusted R^2 (%)	10.17	18.40	0.59	5.13
Number of observations	8962	7314	9682	5796

The table presents results from panel regressions where 12-month log changes in fundamental variables (retail sales, inverse of unemployment, and consumer and producer prices) are regressed on the contemporaneous 12-month log change of industrial production. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include currency fixed effects, but they are not reported. The sample period is January 1976 to March 2017.

Appendix Table 4: Return correlations between trend strategies

	Economic activity	Inflation	Short term	Medium term	Long term
Economic activity	1.000				
Inflation	0.206	1.000			
Short term			1.000		
Medium term			0.812	1.000	
Long term			0.640	0.870	1.000

The table presents correlation coefficients between monthly returns on the trend strategies. The sample period is January 1976 to March 2017.

Appendix Table 5: Return correlations between trend and benchmark strategies

	Economic activity	Inflation	Combo	Carry	Momentum	Value
Economic activity	1.000	0.206	0.766	0.214	0.066	-0.007
Inflation		1.000	0.774	0.578	0.001	-0.042
Combo			1.000	0.502	0.062	-0.035
Carry				1.000	0.142	-0.057
Momentum					1.000	-0.445
Value						1.000

The table presents correlation coefficients between monthly returns on the trend and benchmark strategies. The sample period is January 1976 to March 2017, but due to the long lookback period for the value benchmark all returns start in February 1981.

Appendix Table 6: Predicting currency returns (January 1976–July 1996)

	Currency returns					
	I	II	III	IV	V	VI
Carry	0.112 (0.020)				0.073 (0.031)	0.057 (0.030)
Momentum		0.101 (0.037)			0.085 (0.046)	0.075 (0.046)
Value			0.071 (0.027)		0.081 (0.035)	0.077 (0.031)
Trend combo				0.117 (0.027)		0.062 (0.028)
Adjusted R^2 (%)	0.41	0.33	0.16	0.46	0.63	0.73
Number of observations	3869	3677	2938	3813	2938	2938

The table presents the results of panel regressions in which next month's currency returns are predicted using the current month's portfolio weights: $R_{c,t+1} = a_t + b\tilde{w}_{c,t} + \epsilon_{c,t+1}$, where $R_{c,t+1}$ denotes the monthly excess return for currency c at time $t + 1$ and $\tilde{w}_{c,t}$ denotes the standardized portfolio weight of currency c at time t . Weights of the trend combo are constructed by weighting all sub-strategies by the inverses of their respective past volatilities, scaling the weights such that they sum to one. Weights are cross-sectionally standardized in each month. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to July 1996, but returns start at different dates due to differences in lookback periods.

Appendix Table 7: Predicting currency returns (August 1996–March 2017)

	Currency returns					
	I	II	III	IV	V	VI
Carry	0.123 (0.039)				0.113 (0.048)	0.064 (0.053)
Momentum		0.036 (0.044)			0.052 (0.051)	0.035 (0.046)
Value			0.125 (0.041)		0.146 (0.041)	0.160 (0.039)
Trend combo				0.148 (0.030)		0.138 (0.035)
Adjusted R^2 (%)	0.31	−0.01	0.32	0.47	0.63	0.96
Number of observations	2663	2603	2603	2659	2555	2555

The table presents the results of panel regressions in which next month's currency returns are predicted using the current month's portfolio weights: $R_{c,t+1} = a_t + b\tilde{w}_{c,t} + \epsilon_{c,t+1}$, where $R_{c,t+1}$ denotes the monthly excess return for currency c at time $t + 1$ and $\tilde{w}_{c,t}$ denotes the standardized portfolio weight of currency c at time t . Weights of the trend combo are constructed by weighting all sub-strategies by the inverses of their respective past volatilities, scaling the weights such that they sum to one. Weights are cross-sectionally standardized in each month. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is August 1996 to March 2017, but returns start at different dates due to differences in lookback periods.

**Appendix Table 8: Predicting currency returns over various forecasting horizons
(January 1976–July 1996)**

	Cumulative currency returns			
	1 month	3 months	6 months	12 months
Carry	0.057 (0.030)	0.015 (0.032)	0.006 (0.037)	−0.010 (0.046)
Momentum	0.075 (0.046)	0.008 (0.036)	−0.021 (0.022)	−0.033 (0.017)
Value	0.077 (0.031)	0.093 (0.034)	0.090 (0.034)	0.084 (0.037)
Trend combo	0.061 (0.028)	0.093 (0.029)	0.102 (0.029)	0.118 (0.030)
Adjusted R^2 (%)	0.73	1.86	3.79	8.05
Number of observations	2938	2890	2842	2746

The table presents the results of panel regressions in which future cumulative currency returns over one-, three-, six-, and 12-month horizons are predicted along with current portfolio weights of three benchmark strategies (i.e., carry, momentum, and value) and the trend combo. The cumulative returns are scaled to a monthly basis to facilitate the comparison of the coefficients. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to July 1996, but returns start at different dates due to differences in lookback periods.

**Appendix Table 9: Predicting currency returns over various forecasting horizons
(August 1996–March 2017)**

	Cumulative currency returns			
	1 month	3 months	6 months	12 months
Carry	0.064 (0.053)	0.039 (0.054)	0.041 (0.055)	0.047 (0.056)
Momentum	0.035 (0.046)	0.040 (0.040)	0.031 (0.033)	−0.001 (0.030)
Value	0.160 (0.039)	0.170 (0.040)	0.165 (0.042)	0.149 (0.037)
Trend combo	0.138 (0.035)	0.148 (0.030)	0.148 (0.036)	0.165 (0.039)
Adjusted R^2 (%)	0.96	3.51	7.11	15.50
Number of observations	2555	2540	2525	2495

The table presents the results of panel regressions in which future cumulative currency returns over one-, three-, six-, and 12-month horizons are predicted along with current portfolio weights of three benchmark strategies (i.e., carry, momentum, and value) and the trend combo. The cumulative returns are scaled to a monthly basis to facilitate the comparison of the coefficients. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is August 1996 to March 2017, but returns start at different dates due to differences in lookback periods.

Appendix Table 10: Predicting dollar depreciation over various forecasting horizons

	Cumulative currency returns			
	1 month	3 months	6 months	12 months
Carry	−0.040 (0.026)	−0.056 (0.027)	−0.056 (0.030)	−0.056 (0.033)
Momentum	0.023 (0.043)	−0.002 (0.028)	−0.019 (0.022)	−0.042 (0.023)
Value	0.108 (0.032)	0.122 (0.032)	0.119 (0.034)	0.108 (0.034)
Trend combo	−0.031 (0.041)	−0.018 (0.040)	−0.015 (0.040)	−0.002 (0.041)
Adjusted R^2 (%)	0.30	1.44	2.91	5.80
Number of observations	5493	5430	5367	5241

The table presents the results of panel regressions in which future cumulative dollar depreciations over one-, three-, six-, and 12-month horizons are predicted along with current portfolio weights of three benchmark strategies (i.e., carry, momentum, and value) and the trend combo. The cumulative returns are scaled to a monthly basis to facilitate the comparison of the coefficients. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

**Appendix Table 11: Predicting dollar depreciation over various forecasting horizons
(January 1976–July 1996)**

	Cumulative currency returns			
	1 month	3 months	6 months	12 months
Carry	−0.063 (0.036)	−0.078 (0.040)	−0.080 (0.042)	−0.091 (0.047)
Momentum	0.030 (0.058)	−0.025 (0.047)	−0.049 (0.035)	−0.061 (0.031)
Value	0.080 (0.039)	0.090 (0.042)	0.084 (0.044)	0.078 (0.045)
Trend combo	−0.129 (0.047)	−0.111 (0.048)	−0.107 (0.047)	−0.098 (0.047)
Adjusted R^2 (%)	0.84	2.81	5.60	10.64
Number of observations	2938	2890	2842	2746

The table presents the results of panel regressions in which future cumulative dollar depreciations over one-, three-, six-, and 12-month horizons are predicted along with current portfolio weights of three benchmark strategies (i.e., carry, momentum, and value) and the trend combo. The cumulative returns are scaled to a monthly basis to facilitate the comparison of the coefficients. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is January 1976 to July 1996, but returns start at different dates due to differences in lookback periods.

**Appendix Table 12: Predicting dollar depreciation over various forecasting horizons
(August 1996–March 2017)**

	Cumulative currency returns			
	1 month	3 months	6 months	12 months
Carry	−0.017 (0.049)	−0.037 (0.046)	−0.032 (0.045)	−0.023 (0.048)
Momentum	0.013 (0.045)	0.015 (0.039)	0.004 (0.032)	−0.030 (0.028)
Value	0.177 (0.042)	0.186 (0.046)	0.179 (0.047)	0.159 (0.040)
Trend combo	0.094 (0.044)	0.098 (0.034)	0.095 (0.035)	0.105 (0.039)
Adjusted R^2 (%)	0.60	2.54	5.07	10.75
Number of observations	2555	2540	2525	2495

The table presents the results of panel regressions in which future cumulative dollar depreciations over one-, three-, six-, and 12-month horizons are predicted along with current portfolio weights of three benchmark strategies (i.e., carry, momentum, and value) and the trend combo. The cumulative returns are scaled to a monthly basis to facilitate the comparison of the coefficients. Point estimates are reported with standard errors, clustered by currency, in parentheses. Adjusted within R^2 values and the number of observations are also reported. The regressions include time fixed effects, though they are not reported. The sample period is August 1996 to March 2017, but returns start at different dates due to differences in lookback periods.

Appendix Table 13: Weight analysis for the momentum strategy and the trend combo

	Momentum		Trend combo		Different signs
	Below median	Above median	Below median	Above median	
Australia	0.398	0.602	0.131	0.869	0.390
Austria	0.618	0.382	0.622	0.378	0.488
Belgium	0.458	0.542	0.696	0.304	0.571
Canada	0.515	0.485	0.462	0.538	0.508
Denmark	0.430	0.570	0.474	0.526	0.510
Eurozone	0.596	0.404	0.727	0.273	0.436
Finland	1.000	0.000	0.750	0.250	0.545
France	0.508	0.492	0.801	0.199	0.546
Germany	0.728	0.272	0.919	0.081	0.272
Ireland	0.460	0.540	0.304	0.696	0.576
Israel	0.366	0.634	0.038	0.962	0.366
Italy	0.326	0.674	0.236	0.764	0.421
Japan	0.594	0.406	0.865	0.135	0.422
Netherlands	0.669	0.331	0.867	0.133	0.404
New Zealand	0.332	0.668	0.541	0.459	0.453
Norway	0.499	0.501	0.336	0.664	0.450
Portugal	0.270	0.730	0.026	0.974	0.270
Spain	0.354	0.646	0.059	0.941	0.346
Sweden	0.516	0.484	0.559	0.441	0.491
Switzerland	0.610	0.390	0.802	0.198	0.388
United Kingdom	0.492	0.508	0.290	0.710	0.445
Average					0.443

The table presents an analysis of the portfolio weights for the momentum benchmark and the trend combo. The first two columns report the fraction of months each currency weight is below or above the monthly median weight for the momentum strategy. By construction, the two columns sum to one. The next two columns report the corresponding fractions for the trend combo. The final column reports the fraction of months the momentum and combo weights for a particular currency have different signs. The last row in that column reports the average fraction across the currencies. The sample period is January 1976 to March 2017.

Appendix Table 14: Weight analysis for the value strategy and the trend combo

	Value		Trend combo		Different signs
	Below median	Above median	Below median	Above median	
Australia	0.717	0.283	0.131	0.869	0.627
Austria	0.707	0.293	0.622	0.378	0.505
Belgium	0.507	0.493	0.696	0.304	0.521
Canada	0.455	0.545	0.462	0.538	0.306
Denmark	0.524	0.476	0.474	0.526	0.467
Eurozone	0.569	0.431	0.727	0.273	0.500
Finland	0.625	0.375	0.750	0.250	0.625
France	0.332	0.668	0.801	0.199	0.707
Germany	0.430	0.570	0.919	0.081	0.593
Ireland	0.330	0.670	0.304	0.696	0.420
Israel	0.637	0.363	0.038	0.962	0.675
Italy	0.693	0.307	0.236	0.764	0.419
Japan	0.551	0.449	0.865	0.135	0.350
Netherlands	0.319	0.681	0.867	0.133	0.705
New Zealand	0.656	0.344	0.541	0.459	0.492
Norway	0.394	0.606	0.336	0.664	0.502
Portugal	0.615	0.385	0.026	0.974	0.606
Spain	0.470	0.530	0.059	0.941	0.470
Sweden	0.271	0.729	0.559	0.441	0.552
Switzerland	0.643	0.357	0.802	0.198	0.411
United Kingdom	0.411	0.589	0.290	0.710	0.559
Average					0.524

The table presents an analysis of the portfolio weights for the value benchmark and the trend combo. The first two columns report the fraction of months each currency weight is below or above the monthly median weight for the value strategy. By construction, the two columns sum to one. The next two columns report the corresponding fractions for the trend combo. The final column reports the fraction of months the value and combo weights for a particular currency have different signs. The last row in that column reports the average fraction across the currencies. The sample period is January 1976 to March 2017.

Appendix Table 15: Volatility timing for the trend strategies

	Returns on combos with volatility timing					
	Economic activity	Economic activity	Inflation	Inflation	Trend combo	Trend combo
Constant	-0.076 (0.378)	-0.601 (0.390)	1.340 (0.569)	1.095 (0.460)	0.613 (0.494)	0.186 (0.477)
Carry		0.027 (0.026)		-0.059 (0.041)		-0.004 (0.036)
Momentum		0.034 (0.023)		0.143 (0.057)		0.073 (0.040)
Value		0.085 (0.023)		0.090 (0.053)		0.059 (0.047)
Economic activity	1.268 (0.059)	1.265 (0.058)				
Inflation			1.045 (0.088)	1.072 (0.083)		
Trend combo					1.524 (0.095)	1.542 (0.097)
Adjusted R^2 (%)	83.93	84.60	80.00	79.18	80.39	80.87
Number of observations	490	434	490	434	490	434

The table presents a contemporaneous regression of the monthly returns of the combos (economic activity, inflation, and trend) with volatility timing on respective combos without volatility timing and the carry, momentum, and value benchmarks. Constants are annualized alphas. Point estimates are reported with Newey and West (1987) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. Adjusted R^2 values and the number of observations are also reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

Appendix Table 16: Volatility timing for the benchmark strategies

	Returns on benchmarks with volatility timing					
	Carry	Carry	Momentum	Momentum	Value	Value
Constant	0.792 (0.349)	0.423 (0.360)	1.516 (0.427)	1.272 (0.358)	-0.245 (0.336)	-0.487 (0.328)
Carry	0.914 (0.050)	0.884 (0.053)		0.059 (0.041)		0.037 (0.037)
Momentum		0.092 (0.032)	0.809 (0.058)	0.805 (0.074)		0.055 (0.045)
Value		0.081 (0.025)		0.051 (0.057)	0.852 (0.050)	0.880 (0.047)
Adjusted R^2 (%)	83.81	83.80	81.22	80.17	82.79	83.18
Number of observations	494	434	482	434	434	434

The table presents a contemporaneous regression of the monthly returns of the benchmarks (carry, momentum, and value) with volatility timing on the benchmarks without volatility timing. Constants are annualized alphas. Point estimates are reported with Newey and West (1987) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. Adjusted R^2 values and the number of observations are also reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

Appendix Table 17: Performance of trend strategies (G10 and emerging markets)

	G10			Emerging markets		
	Economic activity	Inflation	Combo	Economic activity	Inflation	Combo
Mean	1.505	3.424	2.391	1.108	6.308	3.569
Standard deviation	5.133	6.398	4.353	6.058	7.725	4.675
Skewness	-0.035	-0.715	-0.605	-0.370	0.254	-0.118
Excess kurtosis	2.428	3.788	2.481	1.773	2.039	2.394
AR(1)	-0.010	0.127	0.054	0.096	0.136	0.219
Sharpe ratio	0.293	0.535	0.549	0.183	0.817	0.763

The table presents performance measures for sub-combos (economic activity and inflation) and the diversified trend combo for the G10 and emerging market countries. Economic activity combines industrial production, retail sales, and the inverse of unemployment; inflation combines consumer and producer price indices. The strategies use rank-based weights each month, going long currencies with relatively strong trends and shorting currencies with relatively weak trends. Each strategy diversifies across sub-strategies by weighting each sub-strategy by the inverse of its past volatility. The measures are based on monthly returns, but means, standard deviations, and Sharpe ratios are annualized. AR(1) refers to the first-order autocorrelation of returns. The sample period is January 1976 to March 2017 for the G10 and January 1997 to March 2017 for emerging markets.

Appendix Table 18: Performance of trend strategies (sub samples)

	January 1976–July 1996			August 1996–March 2017		
	Economic activity	Inflation	Combo	Economic activity	Inflation	Combo
Mean	1.411	3.015	2.107	1.629	3.536	2.580
Standard deviation	2.996	4.357	2.877	4.629	5.512	3.721
Skewness	−0.163	−0.688	−0.708	0.359	−0.814	−0.631
Excess kurtosis	1.941	2.654	2.816	0.988	5.957	2.503
AR(1)	−0.024	−0.032	−0.057	−0.046	0.120	0.056
Sharpe ratio	0.471	0.692	0.732	0.352	0.642	0.693

The table presents performance measures for sub-combos (economic activity and inflation) and the diversified trend combo for the sub-periods January 1976 to July 1996 and August 1996 to March 2017. Economic activity combines industrial production, retail sales, and the inverse of unemployment; inflation combines consumer and producer price indices. The strategies use rank-based weights each month, going long currencies with relatively strong trends and shorting currencies with relatively weak trends. Each strategy diversifies across sub-strategies by weighting each sub-strategy by the inverse of its past volatility. The measures are based on monthly returns, but means, standard deviations, and Sharpe ratios are annualized. AR(1) refers to the first-order autocorrelation of returns.

Appendix Table 19: Performance of time-series trend strategies (G10, developed markets, and emerging markets)

	G10			Developed markets			Emerging markets		
	Economic activity	Inflation	Combo	Economic activity	Inflation	Combo	Economic activity	Inflation	Combo
Mean	0.566	2.561	1.625	0.300	2.642	1.421	3.961	4.120	4.237
Standard deviation	4.907	3.737	2.778	5.012	3.600	2.394	5.884	4.877	3.382
Skewness	-0.185	1.023	0.340	-0.381	1.505	0.085	-1.452	0.353	1.040
Excess kurtosis	4.117	8.300	6.995	4.468	10.113	6.679	8.610	1.575	4.605
AR(1)	0.079	0.100	-0.039	0.088	0.104	-0.011	0.235	0.062	0.264
Sharpe ratio	0.115	0.685	0.585	0.060	0.734	0.594	0.673	0.845	1.253

The table presents performance measures for sub-combos (economic activity and inflation) and the diversified trend combo for the G10, developed, and emerging market countries. The strategies are time-series strategies, allowing for a dollar exposure. Economic activity combines industrial production, retail sales, and the inverse of unemployment; inflation combines consumer and producer price indices. The strategies use time-series weights each month, going long (short) currencies with trends stronger (weaker) than that of the base currency, the USD. Each strategy diversifies across sub-strategies by weighting each sub-strategy by the inverse of its past volatility. The measures are based on monthly returns, but means, standard deviations, and Sharpe ratios are annualized. AR(1) refers to the first-order autocorrelation of returns. The sample period is January 1976 to March 2017 for the G10 and developed markets, and January 1997 to March 2017 for emerging markets.

Appendix Table 20: Factor regressions for the time-series trend combo

	Returns on trend combo		
	I	II	III
Constant	1.030 (0.372)	2.203 (0.499)	1.087 (0.395)
Carry	0.249 (0.035)		
Momentum	-0.012 (0.024)		
Value	0.084 (0.020)		
Dollar carry	-0.036 (0.025)		
Δ FXVOL		-0.002 (0.003)	
Δ TED		-0.002 (0.001)	
Δ VIX		-0.009 (0.010)	
FMP FXVOL			-0.079 (0.031)
FMP TED			-0.046 (0.012)
FMP VIX			0.002 (0.001)
Adjusted R^2 (%)	23.56	0.19	5.71
Number of observations	434	326	490

The table presents a contemporaneous regression of the monthly returns of the trend combo on benchmark strategies and changes in measures of volatility and funding conditions. The trend combo and benchmark strategies use time-series weights, allowing for a dollar exposure. Specification I uses the carry, momentum, value, and dollar carry benchmarks. Specification II uses changes in foreign exchange volatility (FXVOL), the TED spread, and the VIX volatility index, which are non-traded factors. Specification III uses factor mimicking portfolios, which are constructed from regressions of the non-traded factors on five carry-sorted portfolios as in Menkhoff et al. (2012a) for the longest available sample of factors. The factors are then mimicked on longest sample period possible. Constants in specifications I and III are annualized alphas, whereas the constant in specification II cannot be interpreted as alpha as the factors are non-traded. Point estimates are reported with Newey and West (1987) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. Adjusted R^2 values and the number of observations are also reported. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods. Specification II starts in January 1990 due to data availability for the VIX volatility index.

Appendix Table 21: Performance of time-series benchmark strategies

	Carry	Momentum	Value	Dollar carry
Mean	3.989	3.278	−1.074	5.511
Standard deviation	5.186	7.804	7.583	8.815
Skewness	−0.523	−0.385	0.647	−0.021
Excess kurtosis	4.959	1.896	2.317	0.726
AR(1)	0.072	0.040	0.079	−0.030
Sharpe ratio	0.769	0.420	−0.142	0.625

The table presents performance measures for carry, momentum, value, and dollar carry strategies. The strategies are time-series strategies, allowing for a dollar exposure. The measures are based on monthly returns, but means, standard deviations, and Sharpe ratios are annualized. AR(1) refers to the first-order autocorrelation of returns. The sample period is January 1976 to March 2017, but returns start at different dates due to differences in lookback periods.

Appendix Table 22: Return correlations between cross-sectional and time-series trend strategies

	Cross-sectional			Time-series		
	Economic activity	Inflation	Combo	Economic activity	Inflation	Combo
<u>Cross-sectional</u>						
Economic activity	1.000	0.206	0.753	0.365	-0.013	0.313
Inflation		1.000	0.789	0.321	0.117	0.439
Combo			1.000	0.435	0.073	0.484
<u>Time-series</u>						
Economic activity				1.000	-0.191	0.682
Inflation					1.000	0.536
Combo						1.000

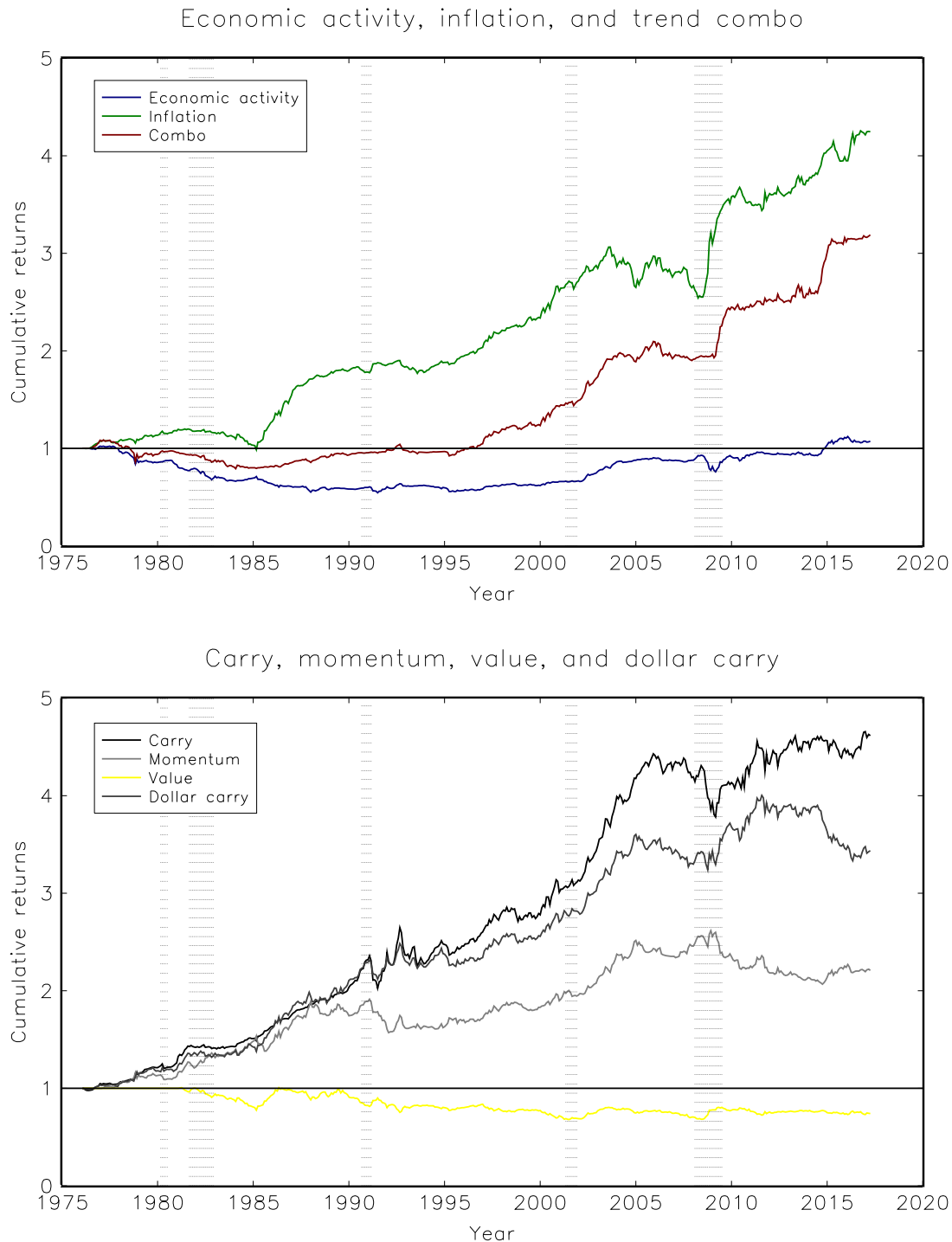
The table presents correlation coefficients between monthly returns on cross-sectional and time-series trend strategies. The sample period is January 1976 to March 2017.

Appendix Table 23: Relating the cross-sectional and time-series combos

	1976–2017		1976–1996		1996–2017	
	Cross-sectional	Time-series	Cross-sectional	Time-series	Cross-sectional	Time-series
Constant	1.389 (0.497)	0.604 (0.403)	2.065 (0.512)	−0.962 (0.465)	0.834 (0.831)	2.067 (0.575)
Cross-sectional		0.348 (0.045)		0.483 (0.070)		0.266 (0.045)
Time-series	0.673 (0.093)		0.746 (0.170)		0.634 (0.109)	
Adjusted R^2 (%)	23.30	23.30	35.77	35.77	16.53	16.53
Number of observations	490	490	242	242	248	248

The table presents contemporaneous regressions in which monthly returns of the cross-sectional combo is regressed onto the time-series combo and vice versa. Constants are annualized alphas. Point estimates are reported with Newey and West (1987) standard errors, accounting for conditional heteroscedasticity and serial correlation up to three lags, in parentheses. Adjusted R^2 values and the number of observations are also reported. The results are reported for the full sample period January 1976 to March 2017, and the two sub-sample periods January 1976 to July 1996 and August 1996 to March 2017.

Appendix Figure 1: Cumulative portfolio returns of time-series trend strategies



The figure shows the cumulative returns of time-series strategies based on fundamental variables. All portfolio returns are scaled ex post as to achieve an annualized ex post volatility of 5%. The color scheme in the top figure is: economic activity (blue), inflation (green), and trend combo (red); the color scheme in the bottom figure is: carry (black), momentum (gray), value (yellow), and dollar carry (dark grey). The shaded areas indicate US contractions (peak to trough) as dated by the NBER. The sample period is January 1976 to March 2017 but cumulative returns start at different dates due to differences in lookback periods.